

TinyLLaMA-based Expert System for IoT Device Security in Application-Specific Domains

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Abstract—The rapid development of Internet of Things (IoT) equipment in domains such as smart homes, agriculture, healthcare services, and industry has intensified concerns about security weaknesses and context-aware safety strategies. This article introduces a lightweight AI-based expert assistant Fine-tuning TinyLlama model, which recommends appropriate authentication methods, identifies possible attacks, and adapts the authentication Method to specific IoT applications and Device contexts. The system operates as a domain-specific advisory tool, rather than directly enforcing security measures in the field, which enables flexible integration into various IoT environments. According to the application and device parameters, an accurate context-driven response was produced using a custom-built dataset and a fine-tuning process. Performance evaluation demonstrated that the system learned effectively, producing high-quality predictions with a final perplexity of 2.41 and accuracy of 85.71%. The suggested assistant's lightweight design makes it especially suitable for deployment in conditions with limited resources while maintaining flexibility in different IoT domains. The results demonstrate how it can be integrated by integrating AI models in IoT security infrastructure.

Index Terms—IoT Security, TinyLLaMA, Device Authentication, Attack Detection, Protection Mechanisms, Smart Home, Smart Agriculture, Healthcare IoT, Industrial IoT, AI-based Expert System, Domain Specific Assistant

I. INTRODUCTION

The rapid proliferation of Internet of Things (IoT) devices across domains such as smart homes, agriculture, healthcare, and industry has brought significant benefits and equally significant security challenges.

Because these devices frequently operate on hardware with limited resources, traditional security solutions are not feasible. Consequently, there is a greater demand for security guidance systems that are domain-specific, lightweight, and flexible. As a way to minimize changing threats and suggest context-aware protection measures, recent research has investigated the integration of Large Language Models (LLMs) into IoT contexts. For instance, leveraging datasets like IoT 23 and TON IoT, an LLM-based framework for automated mitigation techniques and real-time threat detection tailored for resource-constrained environments was developed [1]. Utilizing Retrieval Augmented Generation,

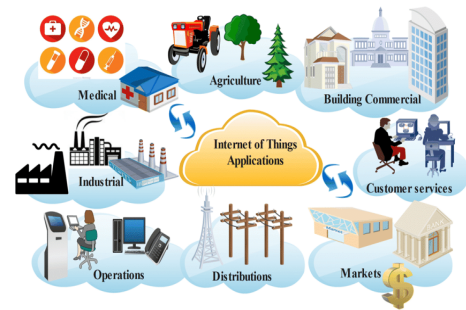


Fig. 1: IoT Applications [23]

ChatIoT, another contemporary system, combines LLM-based reasoning with developing security knowledge, demonstrating notable increases in technical depth and relevance compared to baseline models. The future potential of Retrieval Augmented Generation (RAG) is demonstrated by ChatIoT [2], another recent development that combines dynamic IoT security information with LLMs such as LLaMA3 and GPT 4o. In comparison to conventional LLMs, ChatIoT improves technical correctness and contextual relevance by more than 10% by transforming a variety of datasets into retrievable formats and matching output with user-specific expertise.

Building on these concepts, our study introduces an expert assistant based on TinyLLaMA that provides authentication methods, analyzes potential attacks, and proposes Protection mechanisms proposes defenses specific to various IoT application areas. Our system serves as an advisor, offering domain-specific recommendations in response to particular structured inputs. The system achieves 85.71% accuracy by fine-tuning TinyLLaMA on a specific dataset of IoT device–security mappings application, Smart Home, Agriculture, Healthcare, and Industry scenarios. These findings show usefulness for users looking for quick, domain-aware security advice in resource-constrained contexts.

A. Motivation

The increasing integration of Internet of Things (IoT) devices into critical domains such as smart homes, agriculture, healthcare, and industrial automation has revolutionized operational efficiency, real-time monitoring, and intelligent decision making. However, that rapid adoption also increased security vulnerabilities because traditional cryptography and authentication techniques frequently use more resources than edge devices can handle or are unable to contextually adjust to threats unique to a given application [20]. Along with this, AI-driven security systems offer significant potential have potential by enabling resource-efficient, flexible, and real-time embedded threat detection and anomaly analysis suitable for heterogeneous IoT environments [21]. Along with this, AI-driven security systems have the potential to provide resource-constrained, flexible, real-time embedded threat detection and anomaly analysis.

B. Objective

- To design and develop a lightweight, domain-specific expert assistant capable of recommending security measures for IoT devices and applications across diverse domains such as smart homes, agriculture, healthcare, and industry.
- To implement a context-aware recommendation mechanism that can suggest authentication methods, identify potential attack vectors, and propose protection mechanisms based on structured user input.
- To fine-tune the TinyLLaMA model for domain-specific reasoning using a custom-created IoT dataset, ensuring high predictive accuracy while remaining suitable for resource-constrained environments.

C. Contribution

- The main contribution first, We develop a lightweight, domain-specific AI expert assistant based on the TinyLlama model to recommend IoT security measures for particular devices and applications in industries like smart homes, agriculture, healthcare, and industry, using a custom-built IoT Dataset.
- After receiving structured input, the algorithm makes intelligent recommendations, such as
 - Possible Attacks
 - Possible Mechanism
 - Authentication Method
- Fine-Tuning TinyLLaMA model for Domain Specific Reasoning, enabling high-quality security recommendations in resource-constrained applications.
- Finally, we design an interactive, Gradio-based user interface that allows real-time engagement with the assistant.

D. Structure of the Research Paper

We give a structural summary of our research paper in this section. The diagram shows the organization of each sector, as seen in Figure 2.

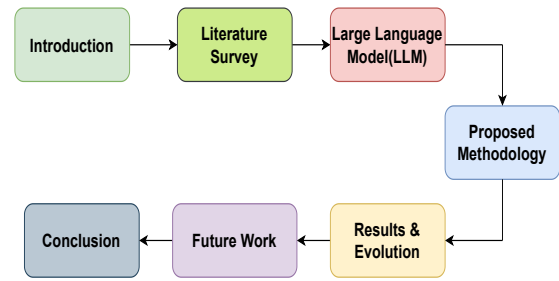


Fig. 2: Schematic representation of the paper's organization.

II. LITERATURE SURVEY

- IoT Security in Diverse Application Domains The proliferation of IoT across smart homes, agriculture, healthcare, and industrial systems has led to tailored threat landscapes in each domain [3]. Table I presents Exploring LLM-Based Intelligence in IoT Applications: Domain-Specific Capabilities and Benefits.

TABLE I: Exploring LLM-Based Intelligence in IoT Applications: Domain-Specific Capabilities and Benefits

Cite	IoT Domain	Specific Application Example	LLM Role / Capability Leveraged	Key Benefits / Challenges Addressed
[4], [5], [6]	Smart Homes	Interpreting user commands, controlling devices, automation	Natural language processing, commonsense reasoning, planning, grounded execution	Context aware control, personalized automation, handling under specified requests
[7]	Agriculture	Precision crop management, livestock monitoring	Data analysis (sensors, imagery), decision support	Optimized resource use, improved yields, sustainable practices
[8], [9], [10]	Industry (IIoT)	Process optimization, predictive maintenance, operational data analysis	Reasoning, reinforcement learning integration, complex data interpretation	Enhanced automation, intelligent decision making, adaptive control
[9], [11], [12], [13], [14]	Healthcare	Patient monitoring, medical query answering, report summarization	Data interpretation (wearable, sensors), plain language generation, knowledge retrieval	Personalized assistance, clinical decision support, automated reporting

Recent advancements in Large Language Models (LLMs) have led to their application across various IoT domains to provide intelligent recommendation systems. Table II presents a comparison of existing LLM-based IoT recommendation systems across four key application areas: smart homes, healthcare, agriculture, and indus-

tries. These systems make use of various datasets and customize datasets also.

TABLE II: Overview of Existing LLM Based Recommendation Systems Across IoT Domains

Cite	IoT Domain	Dataset Used	Recommendation	Method
[15]	Smart Home	RecipeNLG	personalized meal recommendations	RecipeBERT, a BERT based NLP
[16]	Smart Healthcare	Customize dataset(Health Food Recommendation)	Text Based	Mixture of Experts (MoE), Large Language Models (LLMs)
[?]	Smart Agriculture	Synthetic pest sampling, Object Detection dataset	Large Language Model(LLM)	synthetic pest sampling, simulate expert reasoning and provide actionable suggestions
[18]	Smart Industries	Customize Dataset	Cyber Physical Recommendation System	LLM, BERT

III. LARGE LANGUAGE MODEL

Large Language Models (LLMs) represent a significant advancement in artificial intelligence, particularly in the realm of natural language processing. These models, typically built upon deep learning architectures like transformers, have the capability to process and generate human like text with remarkable fluency and coherence. LLMs are trained on vast amounts of text data, learning the intricate patterns, semantics, and nuances of human language. These developments have made it possible for LLMs to be used in a wide range of industries and applications. From conversational agents and natural language comprehension to LLMs have proven their adaptability and usefulness in a variety of fields, including text production and sentiment analysis. Some of the most widely used LLM applications are:

- Natural Language Understanding and Conversational Skills: Proficient in comprehending and producing text that resembles that of a human, enabling applications like virtual chatbots, language translation services, and assistants [18].
- Text Generation: LLMs are useful for content generation, narrative, and creative writing because they can produce text that is logical and pertinent to the context [18].
- Sentiment analysis and information retrieval are made easier by LLMs' ability to sift through large amounts of text data and retrieve relevant information [18].
- Code development is the capacity to create programs or bits of code in languages like PHP, Python, and JavaScript using natural language interpret descriptions, supporting programming and software development activities [18].

IV. PROPOSED METHODOLOGY

A. System Overview

The proposed system utilizes a lightweight large language model (TinyLLaMA) that has been fine tuned using a custom built dataset containing IoT device contexts and security recommendations across multiple domains. The architecture begins with a user input interface, where key contextual information about the IoT application and device is submitted. This input is first preprocessed, including tokenization and

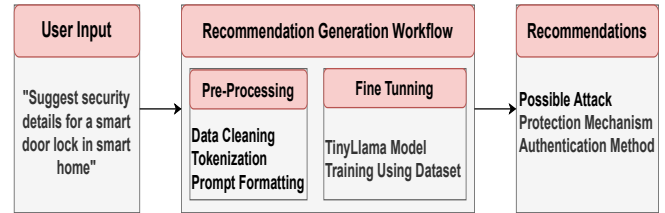


Fig. 3: Block diagram of expert system

prompt structuring compatible with the model format. The processed input is then passed to the fine-tuned TinyLLaMA model, which generates intelligent recommendations based on previously learned patterns in the domain-specific dataset.

- Possible attacks
- Protection Mechanism
- Authentication method

Finally, the entire system is visualized and deployed using Gradio, enabling researchers and users to interact with the model in real time through a shareable link, making the assistant easily accessible and testable from any device.

B. Dataset Creation

We created a customized dataset for IoT applications in four main domains: Smart Home, Agriculture, Industry, and Healthcare. This allowed our domain-specific expert assistant to produce accurate, context-aware recommendations.

1) Dataset Structure:

Input:
"Suggest security details for a Heart Rate Monitor in Healthcare"

Response:
[Model output]
Possible Attacks
Protections Mechanisms
Authentication Method

- Each data entry in the dataset is composed of the following fields
 - Instruction: A natural language prompt describing the application domain, device type, and a specific security concern.

- Output: A concise recommendation containing the suggested authentication methods, potential threats, and appropriate protection techniques.
- Metadata: Tags for domain classification.

2) Data Collection:

- Data was curated from:
 - The work also involves the use of web scraping, along with references to existing IoT security frameworks, public datasets, URLs, and device datasets.

C. Mathematical Model

In this section, a simplified mathematical model is presented to formalize the operation of the proposed TinyLLaMA-based IoT security assistant. The objective is to explain the model's output prediction process, how structured IoT information is handled, and how these predictions are translated into useful security advice.

- Input Representation:
Let:

$$X = \{x_1, x_2, \dots, x_n\}$$

be the input features describing the IoT device, application type, and operational context.

- Training Objective:
We minimize the cross entropy loss:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log P(w_i | w_{<i}, X)$$

where N is the number of tokens in the training sequence.

- Recommendation Mapping:
The output is structured as:

$$R = \{\text{Attack_Type}, \text{Protection_Mechanism}, \text{Authentication_Method}\} \quad (1)$$

where R represents the final recommendation set produced by the system.

D. Preprocessing Pipeline

- To prepare the data for fine-tuning the TinyLLaMA model, the following preprocessing steps were applied:
 - Cleaning: Removal of noisy or duplicate entries.
 - Prompt Formatting: Each input was converted into a consistent **Instruction** → **Response** format.
 - Tokenization: As a way to preserve constant sequence lengths, inputs and outputs were tokenized using the LLaMA-compatible tokenizer, which also performed padding and truncation.
 - Splitting: To validate the model and assess its accuracy, the dataset was divided into 80% training and 20% testing sets.

V. RESULTS AND EVOLUTION

A. Training Parameters

The performance of the TinyLLaMA-based IoT security assistant was optimized for domain-specific security suggestions using the customized IoT dataset. A system with an NVIDIA RTX 3060 GPU (12 GB VRAM) and 32 GB RAM was used for the training procedure.

The essential hyperparameter needed for fine-tuning are compiled in Table III.

TABLE III: Training Parameters for TinyLLaMA Fine Tuning

Parameter	Value
Learning rate	2e-5
Batch size	16
Number of epochs	57
Optimizer	AdamW
Loss function	Cross-Entropy Loss

B. Evaluation Metrics

To assess the performance of the proposed TinyLLaMA-based IoT security assistant, perplexity (PP) was employed as the primary evaluation metric. Perplexity is a standard measure in language modeling that indicates how well a model predicts a sequence of tokens. Mathematically, it is defined as:

- Mathematically, the short form of perplexity is given by:

$$PP = e^{\mathcal{L}}$$

where $\mathcal{L}$ denotes the cross entropy loss between the predicted probability distribution and the true target distribution.

- The full expanded form is expressed as:

$$PP = \exp \left(-\frac{1}{N} \sum_{i=1}^N \log P(w_i | w_{1:i-1}) \right)$$

where:

- N is the total number of tokens in the evaluation sequence,
- $P(w_i | w_{1:i-1})$ is the predicted probability of the i^{th} token given all previous tokens.

In our experiments, the proposed model achieved a final perplexity of 2.41, demonstrating high-quality predictions in generating security recommendations for IoT devices and applications.

C. Quantitative Results

The performance of the proposed TinyLLaMA-based IoT security assistant was quantitatively evaluated using the perplexity metric, which measures the model's ability to predict the next token in a sequence. Perplexity is mathematically defined as:

$$PP = e^{\mathcal{L}}$$

where $\mathcal{L}$ is the cross entropy loss. Lower perplexity values indicate better predictive accuracy.

Table IV presents the final perplexity score obtained after fine-tuning on the custom IoT dataset. The proposed model achieved a perplexity of 2.41, which demonstrates high-quality predictions and effective learning for security recommendation tasks. This low perplexity score reflects the model's ability to capture domain-specific patterns in IoT device and application contexts.

TABLE IV: Perplexity and Accuracy Results on IoT Security Recommendation Task

Model	Perplexity	Accuracy
TinyLLaMA	2.41	85.71%

1) *Performance Evaluation*: As shown in Figure 4, the training loss decreases significantly over epochs, eventually stabilizing at 0.88. The corresponding perplexity also drops steadily and converges to 2.41, demonstrating effective model learning on our dataset.



Fig. 4: Training loss and perplexity curves for the fine-tuned TinyLLaMA model. Final training loss = 0.88, Perplexity = 2.41.

D. Qualitative Results

We provide illustrative examples from several important IoT application categories to illustrate the usefulness of our solution in real-world scenarios. Figures 3 to 6 show detailed visualizations of the recommendation outputs for each domain: Smart Home 5a, Agriculture 5b, Industry 5c, Healthcare 5d.

- Suggest security details for a Heart Rate Monitor in Healthcare.
- Suggest security details for a Smart Door Lock in a Smart Home.
- Suggest security details for a Soil Moisture Sensor in Smart Agriculture.
- Suggest security details for an industrial Robot Arm in industries IoT

A Gradio-based interface was implemented to enable real-time interaction with the model [22].

subcaption

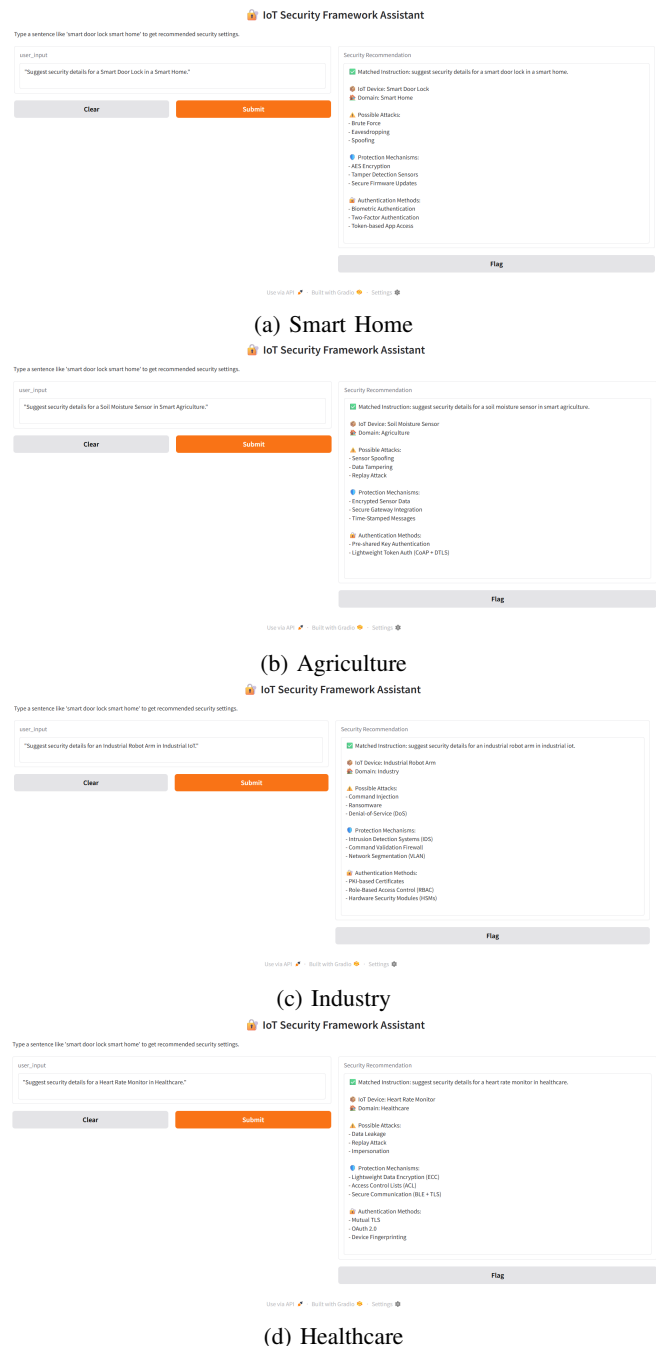


Fig. 5: Recommendation outputs for each IoT domain.

VI. FUTURE WORK

In future iterations, the scope of the proposed system will be expanded to include a broader range of IoT application domains such as Smart Grid, Smart City, Smart Transportation, and Smart Retail. This will be supported by the systematic collection of structured inputs and expert-validated outputs, enabling the creation of a more diverse and representative domain-specific dataset. Additional input parameters that cover not only the application type and device specs but also particular security concerns, operational requirements, and different levels of data sensitivity will be added to im-

prove flexibility and adaptability. A wider variety of technical recommendations will be provided by the expanded system on the output side. These recommendations will cover hardware configurations, software platforms, computing environments, operating systems, communication protocols, authentication methods, device-level and network-level security measures, and circuit and chip-level security improvements. Furthermore, the recommendations will include operating systems optimized for embedded IoT devices, enabling resource-efficient and secure deployments.

VII. CONCLUSION

In this study, we developed a domain-specific, lightweight Large Language Model fine-tuned to provide IoT security framework recommendations. Fine-tuning on customized datasets that integrate four critical IoT domains: smart home, smart agriculture, smart industry, and healthcare. The model can generate context-aware and application-specific security recommendations. The effectiveness of the model was shown by the performance evaluation, which obtained a final training loss of 0.88 and a low perplexity of 2.41, and an accuracy is 85.71%, which reflects strong predictive accuracy and advanced language comprehension. Despite promising results, the study has certain limitations. Assessment was conducted on a limited data set containing only selected IoT applications, such as smart home, health care, industry, and agriculture. The system's lightweight design makes it ideal for implementation in IoT scenarios with limited resources, providing intelligent, real-time recommendations without requiring a large amount of processing power.

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